Introduction to Fluid Mechanics
ME 251

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Course Objectives

- Understand the Characteristics/Properties of fluids
- Have good knowledge of fluid statics
  - ✓ Static Pressure in Fluid
  - ✓ Manometers
  - ✓ Hydrostatic forces
  - ✓ Buoyancy and flotation
- Appreciate the importance of dimensions and dimensional analysis in problem solving.

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Areas to Cover

- **Characteristics/Properties of Fluid**
  - ✓ Mass
  - ✓ Density
  - ✓ Viscosity
  - ✓ Surface tension
  - ✓ Capillarity
  - ✓ Compressibility
  - ✓ Thermodynamic properties (internal energy, enthalpy, etc.)

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Areas to Cover

- **Fluid Statics**
  - ✓ Pressure variation in a fluid at rest
  - ✓ Absolute, gauge, atmospheric and vacuum pressures
  - ✓ Measurement of pressure: Manometers, mechanical gauges
  - ✓ Hydrostatic forces on submerged surfaces
  - ✓ Buoyancy and flotation

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Areas to Cover

- **Kinematics of Fluid Flow**
  - ✓ Methods of describing fluid Motion
  - ✓ Types of fluid flow;
  - ✓ Rate of flow/discharge;
  - ✓ Continuity equation;
  - ✓ Velocity and acceleration.
- **Ideal fluid flow:**
  - ✓ Velocity potential function and stream function.

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Areas to Cover

- **Introduction to Dimensional Analysis**
  - ✓ Secondary or derived quantities,
  - ✓ Dimensional homogeneity (key requirement of valid equations),
  - ✓ Methods of dimensional analysis:
    - Raleigh Method
    - Buckingham's pi theorem

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Instruction Format

Lecture Schedule

Automobile	Eng.	Tuesday	VCR	10.30 – 12.25
Marine	Eng.	√	N2	13:00 – 14:55
Indust.	Eng.	Friday	LT	08:00 – 09:55

Laboratory sessions will be taken in ME 295

Mid- Semester Exam

xxx

End- Semester Exam

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Assessment

- Random and Unannounced Quizzes
- Assignments
- Attendance
- Mid Semester Examination
- End of Semester Examination (70%)

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Useful Reference Books

1. Cengel Y.A. and Cimbala J.M (2006) Fluid Mechaics: Fundamentals and Application, McGraw-Hill Companies Inc
2. Cengel, Y. A., & Turner, R. A. (2001). *Fundamentals of Thermal-Fluid Sciences*. New York: McGraw-Hill Companies Inc.
3. Crowe, C., & Elger, D. (2009). *Engineering Fluid Mechanics*. John Wiley & Sons, Inc.
4. Fox, R. W., McDonald, A. T., & Pritchard, P. J. (2004). *Introduction to Fluid Mechanics* (Sixth Edition ed.). Bogota: John Wiley & Sons, Inc.
5. NAKAYAMA, Y. (2000). *Introduction to Fluid Mechanics*. Tokyo: Yokendo Co. Ltd.

What is Fluid Mechanics?

- The science that deals with the behaviour of:
  - ✓ Fluids at rest (**fluid statics**) or
  - ✓ Fluid in motion (**fluid dynamics**), and
  - ✓ Interaction of fluids with solids or fluids at the boundaries.

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Areas of Fluid Mechanics

✓ **Hydrodynamics -** Study of the motion of fluids that are practically incompressible (such as liquids, especially water, and gases at low speeds).

- The study that deals with liquid flows in pipes and open channels is referred to as **hydraulics**

✓ **Aerodynamics** deals with the flow of gases (especially air) over bodies such as aircraft, rockets, and automobiles at high or low speeds.

✓ **Gas dynamics** deals with the flow of fluids that undergo significant density changes, such as the flow of gases through nozzles at high speeds.

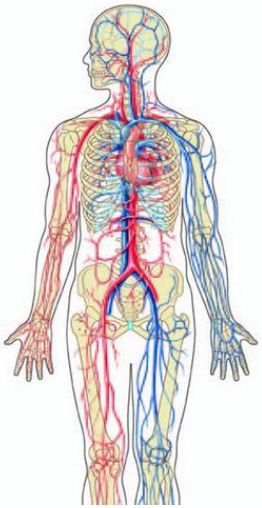
✓ **Meteorology, Oceanography, and Hydrology** deal with naturally occurring flows.

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Application Areas of Fluid Mechanics

- Widely used both in everyday activities and in the desigr of modern engineering systems.
- Fluid mechanics plays a vital role in the human body.
- The heart is constantly pumping blood to all parts of the human body through the arteries and veins, and the lungs are the sites of airflow in alternating directions.
  - All artificial hearts, breathing machines, and dialysis systems are designed using fluid dynamics.





Application Areas of Fluid Mechanics

- ✓ The piping systems for cold water, natural gas, and sewage for an individual house and the entire city.
- ✓ Piping and ducting network of heating and air-conditioning systems.
- ✓ Refrigerators, operation of ordinary faucets (taps), etc.
- ❖ Fluid mechanics plays a major role in the design of all these components.



Application Areas of Fluid Mechanics

- In transportation (Automobiles):

  - ✓ Movement of fuel from the fuel tank to the cylinders—the fuel line, fuel pump, fuel injectors, or carburetors—



  - ✓ Mixing of the fuel and the air in the cylinders

  - ✓ Purging of combustion gases in exhaust pipes are analyzed using fluid mechanics.



Application Areas of Fluid Mechanics

- Fluid mechanics is also used in the:
  - ✓ design of the heating and air-conditioning system,
  - ✓ hydraulic brakes,
  - ✓ power steering,
  - ✓ automatic transmission,
  - ✓ and lubrication systems,



- The shape of recent cars is the result of efforts to minimize drag by using extensive analysis of flow over surfaces.



Application Areas of Fluid Mechanics

- On a broader scale, fluid mechanics plays a major part in the design and analysis of aircraft, boats, submarines, rockets, jet engines, wind turbines, biomedical devices, the cooling of electronic components, and the transportation of water, crude oil, and natural gas.
- In the design of buildings, bridges, and even billboards.
- Numerous natural phenomena such as the rain cycle, weather patterns, the rise of ground water to the top of trees, winds, ocean waves, and currents in large water bodies are also governed by the principles of fluid mechanics.











Characteristics of Fluids

Characteristics of Fluid

Properties of fluid (Density, mass density, viscosity). Thermodynamic properties (compressibility, surface tension and capillarity).

What Is a Fluid?

- Substances may exist in three primary phases:
  - ✓ **Solid**
  - ✓ **Liquid**, and
  - ✓ **Gas**

  } **Fluids**

  - A fourth phase called **plasma** has also been defined for hot ionized gases.





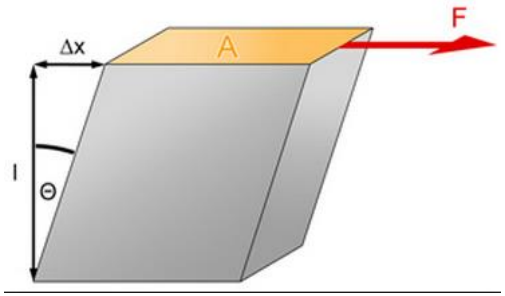
More formally:

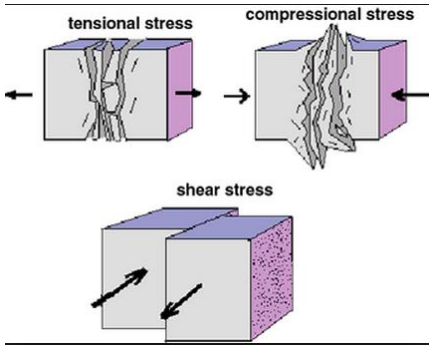
A fluid is substance that can flow or ***continually deforms*** under an applied ***shear stress***.

What is a Fluid?

- Distinction between a solid and a fluid is made on the basis of the substance's ability to resist an applied **shear (or tangential) stress** that tends to change its shape.

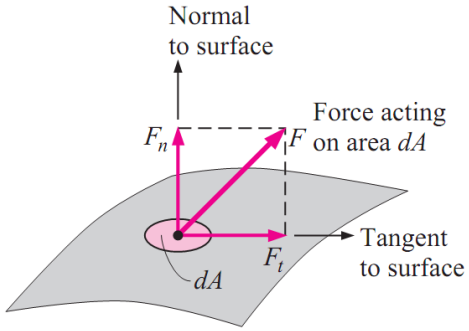
- A solid can resist an applied shear stress by deforming, whereas a fluid **deforms continuously** under the influence of **shear stress**, no matter how small.





What is a Fluid?

- Stress is defined as force per unit area (dividing the force by the area upon which it acts).
- The normal component of the force acting on a surface per unit area is called the **normal stress**.
- The tangential component of a force acting on a surface per unit area is called **shear stress**.



$$\text{Normal stress: } \sigma = \frac{F_n}{dA}$$

$$\text{Shear stress: } \tau = \frac{F_t}{dA}$$

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What is a Fluid?

- In a fluid at rest, the **normal stress** is called **pressure**.

- The supporting walls of a fluid eliminate shear stress, and thus a fluid at rest has zero shear stress.



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Solids and Fluids – A Distinction

- The molecules of a solid are usually closer and the attractive forces between the molecules are so large that a solid tends to retain its shape.
- The intermolecular cohesive forces in a fluid are not great enough to hold the various elements of the fluid together.
- Therefore fluids to flow under the action of the slightest stress and flow will continue as long as the stress is present.

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Solids and Fluids – A Distinction

